Section C will contain extended answer questions on contemporary contexts. This may contain stimulus materials on a scenario that students must read in order to answer the questions. It will focus on the chemistry behind the contexts and will not be a comprehension exercise.

Quality of written communication will be assessed in this examination, in either Section B or C. Questions on the analysis and evaluation of practical work will also be included in either Section B or C.

2.3 Shapes of molecules and ions

Students will be assessed on their ability to:

a demonstrate an understanding of the use of electron-pair repulsion theory to interpret and predict the shapes of simple molecules and ions

b recall and explain the shapes of $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, gaseous PCl_5 and SF_6 and the simple organic molecules listed in Units 1 and 2

c apply the electron-pair repulsion theory to predict the shapes of molecules and ions analogous to those in 2.3b

d demonstrate an understanding of the terms bond length and bond angle and predict approximate bond angles in simple molecules and ions

e discuss the different structures formed by carbon atoms, including graphite, diamond, fullerenes and carbon nanotubes, and the applications of these, eg the potential to use nanotubes as vehicles to carry drugs into cells.